Developing a QR Code-based Library Management System with Case Study of Private School in Surakarta City Indonesia

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***Abstract* -- Manual library management system used in some private schools in Surakarta city has some drawbacks such as it needs longer time for borrowing process, member and collection checking, as well as report preparation. A QR code based computer information system was proposed to solve these problems. This paper discusses the development process of the system and then followed by the tests conducted and discussion. The objective of the work is to record the borrowing, extension, and returning processes as well as their report preparation using QR code based library management system. The information system was built in web-based environment using Bootstrap framework and the data was managed by using MySQL database management system. A camera connected to the computer was used to read the QR code. The test results showed that the optimum distance between QR code tagged book and camera was between 15 and 30 centimetres. The blackbox testing results showed that all functions of information systems working well as expected. The user acceptance test results showed that the system significant acceptance from the respondents so that the information system could be characterized as easy to use, functions and features provided meet the requirement, and able to increase service to the student. Finally, respondents suggested that they would like to implement the system at their school libraries.**

Keywords--QR code, library management system, library information system, school library, web-based system.

I. INTRODUCTION

Managing library manually, i.e. based on pencil and paper, was common practice in many libraries including school libraries. Based on the empirical observation at two private schools in Surakarta City, Indonesia, it can be noted that problems usually arise in manual process in library are including (1) the time spent for borrowing, extension and returning transaction, (2) the difficulty of tracking the books and other loanable resources, (3) the time spent for compiling report, (4) the time spent for checking whether the student has returned the book, and (5) although not significant, the security of the book from the thief. Another key requirement is the implementation cost should be low. Computer-based information systems have benefits to be explored to solve these problems. Managing library using computer-based information system enables library staff to record and track its collection as well as to compile a report [1]. In more advanced development, a library management system, when implementing in the architecture of cloud computing, offer advantages such as it enables to share collection with other institution so that students from other institution able to access the catalogue [2].

Theoritical perspective studies in [3] outlined that Quick Response (QR) code is one of technologies that could be applied in library management system and suggested that the use of QR code for library offer some advantages involving such as fast process, able to store big data, uses standard technology (only camera to percept the code), can be used by user without specific skills so that it would be able to enhance service quality such as to shorten borrowing or returning transaction as well as increasing security of the books from theft in low cost. Based on these advantages, here, in this published study, QR code was used in the development of library management system. The objectives of this published research were: (1) to save the data of the library collection (which are mainly books) in the QR code, (2) to record borrowing, extention, and returning process in the database, (3) to track the students' status of borrowing books. Throughout the text, the development process of the library management system would be presented and would be followed by tests and discussion. The text is pivoted on the practical-empirical point of view of the system development and testing.

II. RELATED WORK

It can be revealed from the study in [4] that Quick Response (QR) codes is two-dimensional code developed by a Japanese company namely Denso-Wave in1994. QR code has capability to store data including 4,296 alphanumeric characters, 7,089 numeric characters, 1,817 Kanji characters and 2,953 binary bytes. Compared to other two dimensional codes such as PDF417, DataMatrix and Maxi codes, QR code has bigger data capacity. Because of its advantages, QR code has been used widely in many areas. Furthermore, research result in [4] showed

that QR code has big potential to be used in education area especially in three subjects including Mathematics, English, and self assesment. QR code also has been proposed in the security area, i.e to protect computer-based health management system [5]. The potential usage of QR code for marketing also has been studied and the results showed that QR code has prospective advantage to increase user satisfaction and marketing efffectiveness [6] [7]. In the area of farming and gardening, Varallyai [8] explored the application of QR code for directing the farm location, and storing information of produced fruits or vegetables. QR codes also could be useful alternative for monitoring attendance of students at the class, and employees at the office or workplaces [9] [10] [11].

The published works by previous researchers showed that there are some technologies could be implemented in the library management system [3], among them are QR code and radio frequency identification (RFID). Xie [12] proposed an RFID-based library management system. This conceptual study showed that the RFID system would be able to enhance both quality of service and security compared to manual library management. However, the drawback of the system is it would be very costly to be implemented in many schools in Indonesia especially in many private schools in Surakarta city because the price of RFID tag for the book is still relatively expensive. In contrast, QR code technology offer lower budget because QR code could be self-generated and then printed by the user. When applied in the library, QR code could be used for many purposes involving directing information of the web, showing the updated time table, showing avalailable electronics resources in the catalogue, downloading document, finding query, and founding bookshelves distribution [13]. QR code also could be used for searching the targetted book to inform its availability and location of the book [14].

III. METHODS

The system development life cycle (SDLC) or waterfall approach [15] was used in the development process. This approach has six serial processes, i.e. requirement analysis, system design, coding, testing, deployment, and monitoring and maintenance.

A. Requirement Analysis

In the development process, requirement analysis was conducted to gain information of what kind of information system that meet the requirement. The analysis was carried out by interviewing librarian and staff combined with direct observation of the services provided in the library. This process resulted conclusion that the information system is expected to have main functions involving: (1) all the borrowing, extension, and returning processes are serviced by library staff or administrator, (2) in order to save time, the staff does not need manually typing the identity of book in the process of borrowing, extention and returning book, (3) the system could track the students' status of borrowing books,whether the student has returned the book or not, (4) the system could compile the report automatically including statement letter that student has returned all books he/she borrowed for administrative usage and print all of them.

The QR code based library management system has three main components: (1) QR code tagged at the book, (2) camera connected to the computer to read the QR code, (3) information systems placed in the computer. The general architecture of the library management system is depicted in Fig. 1.

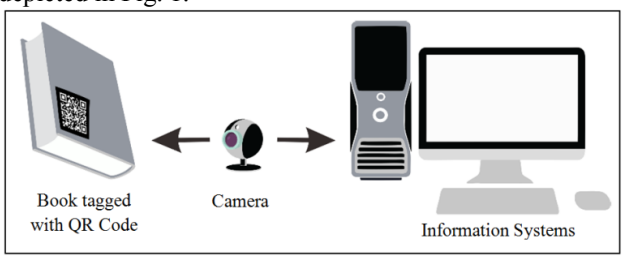

Fig. 1. General system architecture.

By using QR code technology, the information or identity of the book will be encoded in the QR code. Actually the information were initiated by the administrator using information systems set up in the computer and then stored it in the database and in the same time the system generated the QR code. QR code is unique for every book. Every book, even with the same title, has different QR code. The QR code was then printed in the piece of paper and tagged to targetted book. To decode information of the book, firstly the information system will read the QR code using camera and then system will match the QR code read versus the code in the system. After that, if the code is match, the system displays the information or identity of the book otherwise it would not display the information of the book.

Information system has two actors, i.e. administrator and library staf. These two actors have different access level. Administrator has full control of the system while library staff has lower access level as indicated in the usecase diagram depicted in Fig. 2.

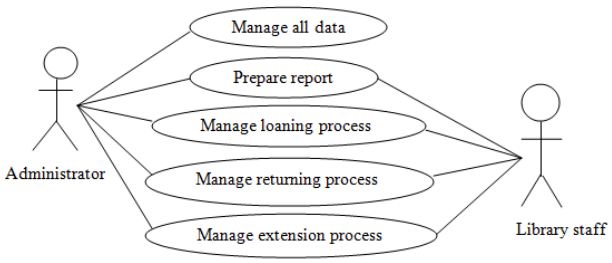

Fig. 2. Usecase diagram of the system

B. Database design

All data are stored in the database. The information system has one database with name *perpusmuha* which has five table, i.e. table for user, *anggota* (student member), *peminjaman* (borrowing process), *det_peminjaman* (detail of borrowing transaction), and *buku* (book or other loanable collections). The five table and the relation between tables is depicted in Fig. 3.

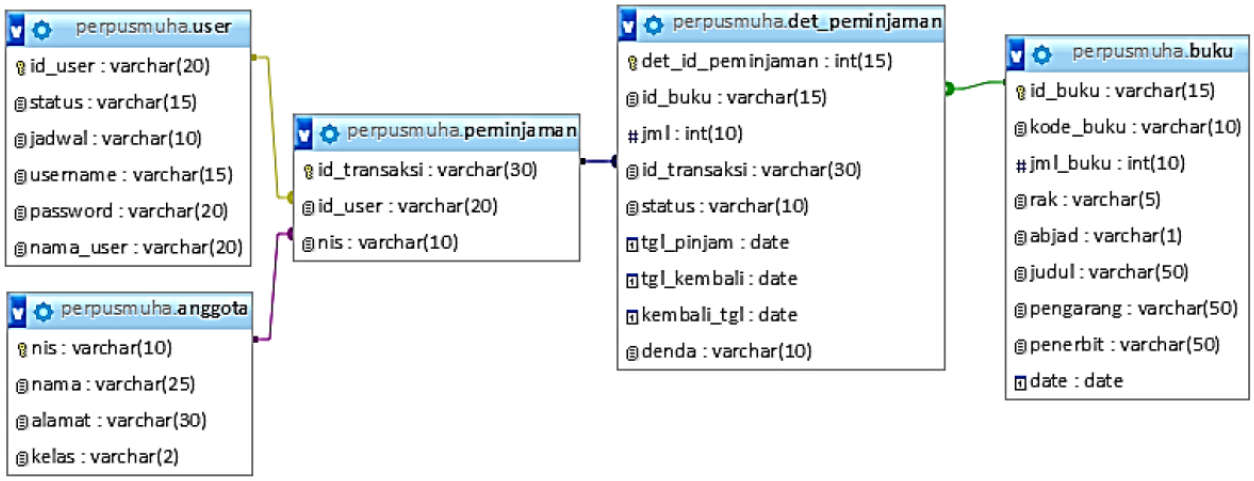

Fig. 3. Tables and their relation in database. (name of table and fields are in Bahasa Indonesia)

C. *Coding process*

The information system was implemented in web-based environment on existing local area network. Some software tools were needed throughout the development process involving: (1) XAMPP version 3.2.1 as a web server anda database server with MySQL serving as the database management systems (DBMS), (2) web browser to display the web-based information system, (3) Sublime Text 3 as the text editor, (4) Bootstrap framework, (5) webcodecamjs plugin for QR code encoding and decoding, (6) datatable for displaying data, (7) datepicker for inputting date, (8) qrcodegen for encrypting primary key code of the QR code, (9) PHP programming language.

IV. RESULTS AND DISCUSSION

A. *Resulted system*

The resulted system and its implementation on personal computer is depicted in Fig. 4 while sample of QR code tagged on the book is depicted in Fig. 5.

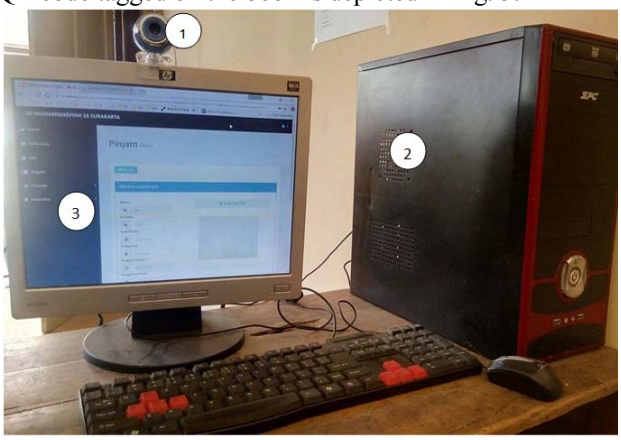

Fig. 4. The implementation of qr code based library management systems: (1) camera connected to the computer to read the QR code, (2) central procesing unit (CPU) of the computer, (3) display of the system.

Fig. 5. The sample of QR code tagged on the book: (1) the area of two dimensional of QR code, (2) the unique number generated for every collection or book.

Most of data and instruction of the system are represented in Bahasa Indonesia. The operational steps of the information systems are summarised as follow. Firstly, the library staff open the system, login to the system, and then choose the menu of transaction which would be processed for example borrowing process (its actual display is depicted in Fig. 6). Then, the library staff just need to choose the student, either name or identity number, and scan the QR code tagged of the book. Afterwards, the system will display the QR code which was scanned by the camera and the data (identity) of the book. The library staff then just need to click the *tambah* (add) button in the bottom of the page. The data for the chosen student and the collection were then updated, i.e. the student has one more book in the borrowing list while in the collection list there was one book taken. The process for extension and returning process also very similar to that of borrowing process.

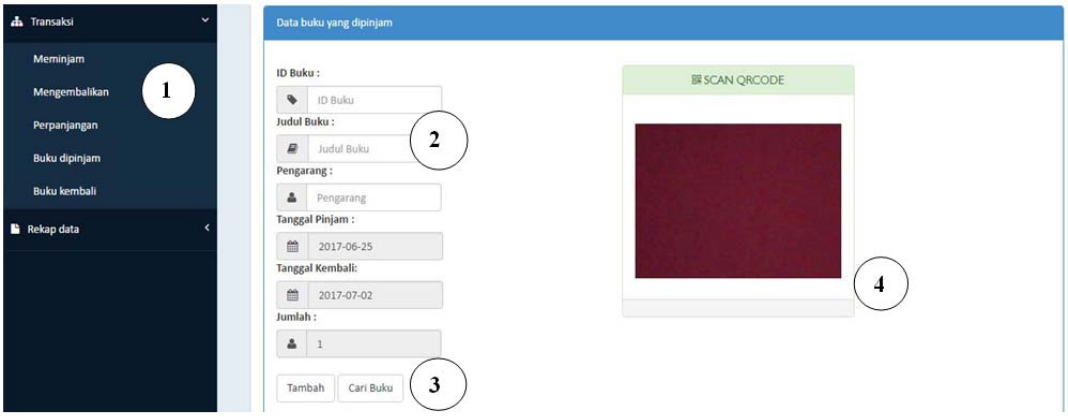

Fig. 6. Actual display of borrowing menu page of the information system: (1) list of transaction menus available in the library management system, (2) data of the book, (3) button for executing the transaction, (4) display of QR code percepted by camera.

Compared to manual system, the QR code based library management system offers simpler process for borrowing, extention, and returning process. The library staff does not need to write manually of the data including identity of the book, the current date and the deadline but just scan the QR code using camera and the data would be recorded in the database automatically. Besides, library staff does not need to bother on the checking the collection manually but just click the recapitulation menu then the system will display a printable file which could be seen in Fig. 7.

No	ID Buku	Kode Buku	Letak Rak	Judul Buku	Penulis	Penerbit	Tahun Terbit	Tanggal Masuk	Jumlah
1	B0000001	11001100	3A	Kejujuran Kunci Keberhasilan	Nurhayati Setiawan	Alfabeta Bandung	2001	2009-12-08	6
2	B0000002	1201345	3A	Bermain dengan matematika	Badrul K., Ruslani	Angkasa Bandung	2007	2009-12-12	8
3	B0000003	1201337	3A	Berain dengan Angka	Badrul K., Ruslani	Angkasa Bandung	2007	2009-12-12	4
4	B0000004	12002456	3A	Anak-anak Pun Dapat Membangun Desa	A.Soerota	Azka Press	2006	2010-01-09	1
5	B0000005	4011896	3A	Teknik Mencangkok Yang Mudah	Suparman	Azka Press	2007	2009-12-12	2
6	B0000006	2502564	3A	Pencuri Semah Kenduri	Abel Tasman	Azka Press	2007	2010-01-09	3
7	B0000007	2002572	3A	Si Kabul Kambing yang setia	E.Sulaiman	Azka Press	2007	2009-12-12	2
8	B0000008	2002484	3A	Nadia & Dukun Geger Kalong	Andarini Trisnasari	Azka Press	2007	2010-01-09	3

Fig. 7. Sample of printable report display: list of library's collection.

B. Experimentation and discussion

1) QR code perception testing

QR code perception testing was carried out to reveal the optimal distance range between QR code tagged book with camera. The test was conducted experimentally inside the standard office room of school library with normal daylight combination of sunlight (from the windows and door) and lamp. The actual lumen was not measured. In the test, firstly, information system was openned in either borrowing, extension, or returning menu. Secondly, QR code tagged book was placed in a distance (d) measured in centimetre to the camera which was connected to the computer. Finally, tester check whether the camera able to read the QR code and system able to display the QR code and read the data or not. Tester continually recorded the d and its results accordingly. The test results presented in Table I showed that the optimum distances are between 15 and 30 centimetres.

TABLE I. DISTANCE BETWEEN QR CODE TAGGED BOOK WITH CAMERA TEST RESULTS

Distance (d)	Results
d<5	Camera able to percept the qr code in a part so that the system could not read the information of the qr code.
5<d<15	Camera able to percept the qr code inconsistently, some time in full some time in a part, so that the system also able read the information of the qr code inconsistently.
15<d<30	Camera able to percept the qr code fully and clearly so that the system could read the information of the qr code consistently.
30<d<40	Camera able to percept the qr code inconsistently, some time able to percept in full some time could not percept, so that the system alsa able read the information of the qr code inconsistently.
d>40	Camera could not able to percept the qr code so that the system could not read the information of the qr code.

The effect of the hindrance of the QR code was experimentally studied by put paper or any solid object

above the QR code tag. Furthermore, a reflective transparent plastic was covered on the QR code tag. The results showed that the system would be able to read the QR code and display the data of the book if the camera could percept the QR code in full without any object cover it. Also, the reflective transparent plastic made system could not read the QR code.

2) Blackbox testing

Blackbox testing [16] was performed to test whether all functions of the information system were working as expected. In the testing process, tester was given a form and then was asked to conduct serial predetermined actions provided in the form and then to record the actual output. The test result was then stated whether the actual output is the same as expected (valid) or not (invalid). In order to assure that the system provided consistent actual output, blackbox testing was carried out many times by many testers. The blackbox testing results summarised in Table II confirmed that all functions are valid.

TABLE II. BLACKBOX TESTING RESULTS

Function	Input	Actual output	Result
Access level control (administrator, user, and unknown)	(i) Tester typed username and password for adminstrator, and user in login page. (ii) Tester typed unknown username and password.	(i) System displayed the main page for administrator and user accordingly. (ii) System denied access.	(i) Valid (ii) Valid
Access control: log out	(i) In the position of logged in, either as administrator or user, tester press log out button. (ii) Then, after successfully logged out, tester press back button of the page.	(i) Tester successfully logged out and system displayed log in page. (ii) Tester could not access previous page.	(i) Valid (ii) Valid
Inputting data	In the logged in position, tester pressed add data button and then input the data by filling the predetermined form.	System recorded and displayed the filled data.	Valid
Editting data	Tester pressed the edit button of targetted data then edited data.	System recorded and displayed the editted data.	Valid
Deletting data	Tester pressed the delete button of targetted data.	System delete the data from the database.	Valid
Borrowing process	Tester chose borrowing menu and then processed the borrowing transaction by using QR code.	System recorded and displayed the borrowed book and marked the book with status “borrowed”.	Valid
Extention process	Tester chose extention menu and then processed the extention transaction by using QR code.	System recorded and displayed the borrowing date.	Valid
Returning process	Tester chose returning menu and then processed the returning transaction by using QR code.	System recorded and displayed the returned book and marked the book with status “returned”.	Valid
Printing collection	Tester openned the collection list page and then pressed print button.	System generated a printable file of collection list and displayed it. Then, printing could be processed by clicking print button.	Valid
Borrowing and returning report	Tester chose report menu and then chose date or range of date of data to be compiled.	System generated a printable file of borrowing and returning data on targetted date and displayed it. Printing could be processed by clicking print button.	Valid
Printing recommendation letter	Tester chose recommendation letter menu and fill student name in the form.	System generated a printable file of recommendation letter of intended student and displayed it. Printing could be processed by clicking print button.	Valid

3) User acceptance test

User acceptance test was performed in the final step to gain user view of the system [16]. Judgement sampling approach [17] was used in the testing. Two head of school libraries who have experience in managing school libraries manually for more than five years were chosen as samples. Both respondents have skill in operating computer including working with web-based systems but have not yet experience operating computer-based library information systems. The user acceptance test was conducting in three main steps. Firstly, researchers introduced the information systems to respondents including all detail functions, buttons and how to operate them. Secondly, respondents were asked to operate the system acting as administrator and library staff. Finally, respondents were asked to fill the predetermined questionnaire. The questionnaire consists of five statements including: (i) P1: the display of the system is simple and easy to understand, (ii) P2: data recording and processing functions meet the requirement, (iii) P3: the system able to increase service of the library to the student, (iv) P4: the system easy to use, (v) P5: the system fulfil the requirement of the library. For each statement, samples were asked to choose one of five available answer alternatives which showing the degree of acceptance involving: (a) strongly agree (score 5), (b) agree (score 4), (c) neutral (score 3), (d) not agree (score 2), and (e) strongly not agree (score 1).

The test results showed that the system gained high acceptance, both respondents chose either strongly agree or agree for every statement. When asked their attitute towards the system, whether they would like to implement

the system for their school libraries, the both respondents said that they would implement it.

4) Discussion

The main advantage of the QR code-based library management system implementation for the school is mainly for saving the time for servicing student and also for preparing report. As an illustration, for servicing time, using QR code-based system staff needs less than 30 seconds either for recording borrowing or returning process per book per student in comparison to between 2 to 3 minutes using manual process. Then, for report preparation, using QR code-based system staff needs less than one minute to compile the report of all the collection status compared to about one working day (between 7 to 8 working hours) using manual process. Another time consumming process was to check whether student has returned all books he or she borrow in the end of every semester and the provide the letter as a requirement for receiving academic report. Using QR code-based system, this task could be completed less than one minute per student compare to about 30 minutes using manual process.

From the fascilities requirement point of view, the implementation of the QR code based system need middle to low specification computer only so that it is not expensive for low-economic level school. Hence, this system is feasible for low-economic level schools, mainly private schools, because it can be noted from field observation that the amount of cost required is the biggest problem for school when they would like to acquire new technology for supporting academic process. Besides budget, another obstacle in the implementation of QR code –based library management system is the human resources. A less complex system will be more suitable than complex system because it will be easier to operate and to maintain.

V. CONCLUSION

Conclusion that could be drawn from the development process, testing, and discussion are involving: (1) QR code has been successfully implemented to store library collection data in the form of tag which can be attached in the library collection, (2) developed information system able to read the data stored in the QR code using camera and using it in the transaction process with optimum distance between 15 and 30 centimetres, (3) blackbox testing results showed that all menus and features of the systems were working well as expected, (4) user acceptance test results showed that the system has high acceptance on its easiness to use, functions and features provided has met the equirement, and able to increase quality of service to the student. All in all, respondents have positive attitutde towards the system and they would like to implement the system at their school libraries. This work could be extended for example using mobile based technology.

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